

QUANTUM STATE REFERENCE: THE UNIFIED MARKET PHYSICS THEORIES

TARGET AUDIENCE: Claude Code (Development AI) **PURPOSE:** This document is the ultimate architectural specification mapping the theoretical physics of the Bayesian-AI trading system to the deterministic state variables and decision execution logic. Use this reference to reconnect the quantum state computations (`bayesian_brain.py` , `physics_utils.py` , `fractal_dmi.py`) to the entry scoring and exit engines (`execution_engine.py` , `core/exits/*`).

1. THEORY I: THE NIGHTMARE PROTOCOL (The Gravitational Field)

The Concept: The market acts as a localized gravitational well. Price is a particle tethered to a moving Center of Mass. The Standard Error bands define the structural integrity of this gravitational field.

1.1 Mathematical Formalization

- **The Ornstein-Uhlenbeck (O-U) Process:** Defines the baseline mean-reverting behavior of the particle.

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

- **The State Position (Z-Score):**

$$Z_t = \frac{X_t - \mu}{\sigma}$$

- **Gravitational Force ($F_{gravity}$):** Pulls the particle back to the mean. It strengthens linearly as distance increases.

$$F_{gravity} = -\theta \cdot Z_t \cdot \sigma$$

- **Boundary Repulsion Force ($F_{repulsion}$):** The electrostatic-like repulsion emitted by the Event Horizon. It scales inversely with the cubed distance to the boundary (r).

$$F_{repulsion} = \frac{k}{r^3} \quad \text{where} \quad r = 3 - |Z_t|$$

- **Roche Limit (2σ):** The boundary where tidal forces begin to tear the particle's localized trend apart.

- **Event Horizon (3σ):** The absolute singularity boundary. Probability density drops to near zero.

1.2 State Variables (Encoding the Theory)

These fields in `MarketState` / `ThreeBodyQuantumState` carry the protocol's state:

- `z_score` : The spatial coordinate of the particle.
- `micro_velocity` : The first derivative of price with respect to time (dP/dt).
- `F_gravity` : The computed magnitude of the restoring force.
- `F_repulsion` : The computed magnitude of the boundary pushback.
- `net_force` : $\sum(F_{gravity}, F_{repulsion}, F_{momentum})$.

1.3 Decision Rules

- **ENTRY (Wave Function Collapse):** Enter only when `net_force` vector aligns with the target state, and price has pulled back to a stable zone ($|Z| \leq 1.0$).
- **EXIT (Roche Limit Hit / Death Hook):** Trigger `fractal_exhaust.py` when price touches 2σ (Roche Limit) AND `micro_velocity` flips sign (kinetic energy reaches zero against the wall).
- **NO-ENTRY (Event Horizon):** Hard block on any trend-continuation entries if $|Z| \geq 2.5$. The particle is too close to the singularity.

2. THEORY II: THE THREE-BODY PROBLEM (Fractal Chaos)

The Concept: The market consists of three interacting gravitational bodies: Macro (1h), Meso (5m), and Micro (15s). Because a three-body system is deterministically chaotic, we must model it probabilistically using quantum wave functions and entropy.

2.1 Mathematical Formalization

- **The Superposition Wave Function (Ψ):** The particle exists in a superposition of three discrete states (Center, Upper Band, Lower Band) until observed/resolved.

$$\Psi = a_0\Psi_{center} + a_1\Psi_{upper} + a_2\Psi_{lower}$$

- **Boltzmann Probabilities (P_i):** The probability of collapsing into state i , calculated via the energy potentials of the Net Force field.

$$P_i = \frac{\exp(-\beta E_i)}{\sum_j \exp(-\beta E_j)}$$

- **Shannon Entropy (S):** Measures the chaos/uncertainty of the three-body system.

$$S = - \sum_i P_i \ln(P_i)$$

- **Fractal Coherence (C):** The dot product (alignment) of the Micro, Meso, and Macro directional vectors.

2.2 State Variables (Encoding the Theory)

These fields carry the quantum state:

- `prob_center` , `prob_upper` , `prob_lower` : The collapsed Boltzmann probabilities (P_i).
- `entropy` : The Shannon entropy score of the current bar.
- `coherence` : The calculated alignment [0.0 to 1.0] of the 3 timeframes.
- `tunnel_probability` : The Monte Carlo-derived probability of the particle tunneling through the current resistance layer.

2.3 Decision Rules

- **STAY FLAT (Chaos):** If `entropy` is HIGH (e.g., probabilities are $\approx [0.33, 0.33, 0.33]$) OR `coherence` is LOW, the three bodies are in chaotic misalignment. Block all entries.
- **ENTER (Resolution):** When `entropy` drops (e.g., $P_{upper} > 0.80$) AND `coherence` is HIGH, the 3-body system has temporarily resolved into a stable orbit. Execute trade in the direction of the highest probability state.
- **EXIT (Decoherence):** Trigger `regime_decay.py` if a previously high `coherence` state rapidly decays into high `entropy` while a trade is open.

3. THEORY III: THE RESONANCE CASCADE (The Alpha Event)

The Concept: Occasionally, the frequencies of the Three Bodies synchronize. The Micro particle does not bounce off the Meso Roche Limit; it resonates with the Macro gravity. The standard statistical boundaries shatter, resulting in violent, exponential directional expansion (a Black Swan / Trend Day).

3.1 Mathematical Formalization

- **The Lyapunov Exponent (λ):** Measures the rate of separation of infinitesimally close trajectories. Determines if the system is stable ($\lambda < 0$) or chaotic/cascading ($\lambda > 0$).

$$|\delta Z(t)| \approx e^{\lambda t} |\delta Z(0)|$$

- **Hurst Exponent (H):** Measures the long-term memory of the series.
 - $H < 0.5$: Mean-reverting (Stable)
 - $H > 0.55$: Persistent (Resonance Cascade)

3.2 State Variables (Encoding the Theory)

- `lyapunov_exponent` : Real-time calculation of λ .
- `macro_hurst_exponent` : Rolling Hurst calculation on the Macro timeframe.
- `is_resonance_cascade` : Boolean flag. True if $H > 0.55$ and $\lambda > 0$.

3.3 Decision Rules

- **HOLD (Cascade Active):** If `is_resonance_cascade == True`, standard physics are suspended. Do not fade the edges.
- **EXIT (Cascade Exhaustion):** Exit cascade mode only when H decays below 0.50 or λ flips negative.

4. THE CASCADE MODE PROTOCOL

When the system detects a true market anomaly, it enters a dedicated execution protocol designed to capture maximum asymmetric upside.

Exact Conditions (`is_resonance_cascade = True`):

1. `macro_hurst_exponent > 0.55`
2. `lyapunov_exponent > 0`
3. `coherence` is near 1.0 (Macro, Meso, and Micro DMI/ADX vectors perfectly aligned).
4. Macro Volatility (σ_{macro}) is expanding directionally.

Suppressed Exits (During Cascade):

- **Standard Take Profit:** Disabled. We do not cap gains during a cascade.
- **Band Exit / Roche Limit Defense:** Disabled. The 2σ and 3σ bands are actively breaking; fading them is mathematically fatal.

Cascade Termination Triggers:

- **Regime Decay (`regime_decay.py`):** The `macro_hurst_exponent` drops below 0.50, signaling a return to mean-reversion physics.
- **Tidal Wave (`tidal_wave.py`):** Volatility suddenly expands violently in the *opposite* direction of the cascade.
- **Vector Collapse:** Macro ADX hooks downward violently.

5. THE ORIGINAL NIGHTMARE FIELD EQUATION (Compute Pipeline)

The complete computation pipeline that transforms raw OHLCV tick data into actionable quantum states.

5.1 The Master Equation

$$dX_t = \underbrace{\theta(\mu(t) - X_t)dt + \sigma dW_t}_{\text{Ornstein-Uhlenbeck}} + \underbrace{\mathcal{F}_{PID}(e)dt}_{\text{Algorithmic Control}} + \underbrace{\mathcal{J}(\lambda)}_{\text{Resonance Jump}}$$

5.2 The Compute Pipeline Step-by-Step

1. Statistical Bounds Computation:

- Ingest raw OHLCV.
- Calculate Linear Regression Mean (μ) and Standard Error Bands (σ) for Macro, Meso, Micro.
- Compute Z -score for current price.

2. Force Field Computation:

- Calculate $F_{gravity} (-\theta \cdot Z \cdot \sigma)$.
- Calculate $F_{repulsion} (\frac{1}{r^3}$ from the 3σ boundary).
- Calculate $F_{momentum}$ (Derivative of the ADX/DMI vector).
- Sum to find F_{net} .

3. Wave Function Computation:

- Convert F_{net} into Potential Energies (E_i) for the three spatial states (Mean, Upper Band, Lower Band).
- Apply Boltzmann weights to compute P_i (prob_center , prob_upper , prob_lower).

4. Tunnel Probability (Monte Carlo):

- Run stochastic O-U path generation projecting forward N bars.
- Calculate the percentage of paths that successfully pierce the localized resistance without breaching the stop-loss boundary.
- Output as `tunnel_probability`.

5. Execution Handoff:

- The `ThreeBodyQuantumState` vector is populated.
- Handoff to `execution_engine.py` and `core/exits/*` to parse the Boolean rules defined in Sections 1-4.